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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2015 Trial Examination

# FORM VI

## MATHEMATICS 2 UNIT

Friday 31st July 2015

### General Instructions

- Reading time — 5 minutes
- Writing time — 3 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

### Total — 100 Marks

- All questions may be attempted.

### Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

### Section II – 90 Marks

- Questions 11–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

### Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

### Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 92 boys

Examiner

DS

**SECTION I - Multiple Choice**

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

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**QUESTION ONE**

**1**

Which expression represents the product of the roots of the equation  $ax^2 + bx + c = 0$ ?

- (A)  $-\frac{c}{a}$
- (B)  $\frac{c}{a}$
- (C)  $-\frac{b}{a}$
- (D)  $\frac{b}{a}$

**QUESTION TWO**

**1**

What is the range of the function  $y = x^2 - 4$ ?

- (A)  $y \geq -4$
- (B)  $y \geq 0$
- (C)  $-2 \leq y \leq 2$
- (D) all real values of  $y$

**QUESTION THREE**

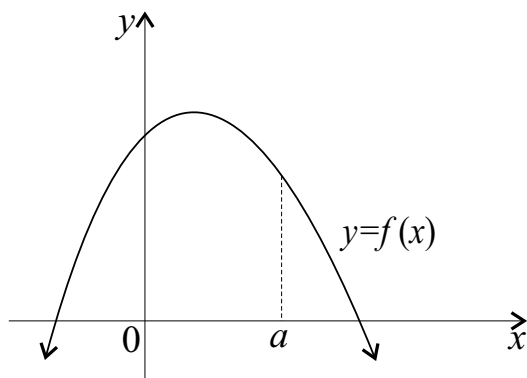
**1**

Which of the following is the derivative of  $x^{n+1}$  with respect to  $x$ ?

- (A)  $nx^{n+1}$
- (B)  $\frac{x^{n+1}}{n+1}$
- (C)  $(n+1)x^n$
- (D)  $(n+1)x^{n-1}$

**QUESTION FOUR**

1



The diagram above shows the graph of the function  $y = f(x)$ .  
Which of the following statements is TRUE?

- (A)  $f(a) < 0$  and  $f'(a) > 0$
- (B)  $f(a) > 0$  and  $f'(a) < 0$
- (C)  $f(a) > 0$  and  $f'(a) > 0$
- (D)  $f(a) < 0$  and  $f'(a) < 0$

**QUESTION FIVE**

1

What are the period and amplitude, respectively, of the wave  $y = \frac{1}{4} \sin 4\pi x$ ?

- (A)  $\frac{\pi}{2}$  and  $\frac{1}{4}$
- (B)  $\frac{1}{4}$  and  $\frac{\pi}{2}$
- (C)  $\frac{1}{2}$  and  $\frac{1}{4}$
- (D)  $\frac{1}{4}$  and  $\frac{1}{2}$

**QUESTION SIX**

1

Which expression is equivalent to  $\log 2a - \log a$ ?

- (A)  $\log 2$
- (B)  $\frac{\log 2a}{\log a}$
- (C)  $\log a$
- (D)  $2 \log a$

**QUESTION SEVEN**

**1**

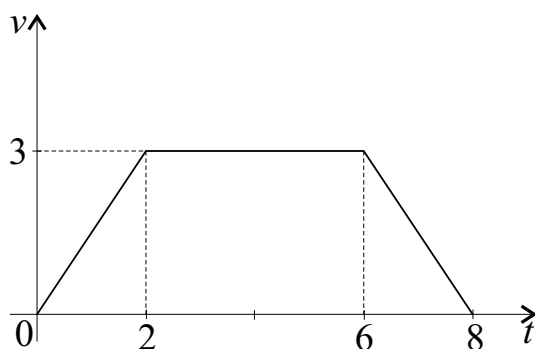
Simpson's rule is used with three function values to approximate  $\int_0^1 9^x dx$ .

What value is obtained?

- (A)  $\frac{25}{12}$
- (B)  $\frac{22}{3}$
- (C)  $\frac{5}{2}$
- (D)  $\frac{11}{3}$

**QUESTION EIGHT**

**1**

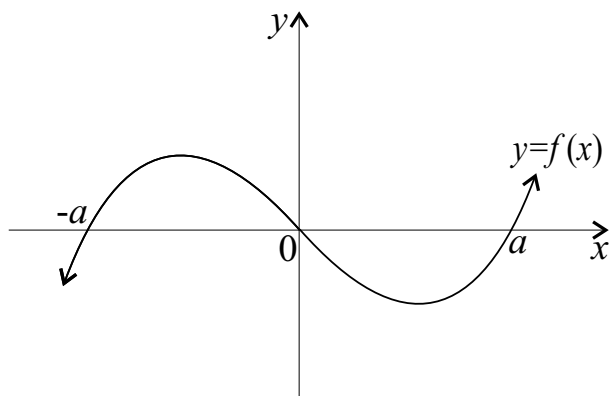


The diagram above shows the velocity–time graph of a particle moving in a straight line. Which of the following statements is TRUE?

- (A) The velocity is constant throughout the motion.
- (B) The particle does not return to its starting point.
- (C) The particle cannot move because its initial velocity is zero.
- (D) The distance travelled by the particle cannot be determined.

**QUESTION NINE**

**1**



The diagram above shows the graph of the ODD function  $y = f(x)$ . Which of the following statements is FALSE?

- (A)  $\int_{-a}^0 f(x) dx > 0$
- (B)  $\int_{-a}^a f(x) dx = 0$
- (C)  $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$
- (D)  $\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$

**QUESTION TEN**

**1**

If  $x$  is a negative number, which of the following statements is FALSE?

- (A)  $|x| = -x$
- (B)  $2x < x$
- (C)  $-x > x$
- (D)  $\sqrt{x^2} = x$

\_\_\_\_\_ End of Section I \_\_\_\_\_

**Exam continues overleaf ...**

**SECTION II - Written Response**

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

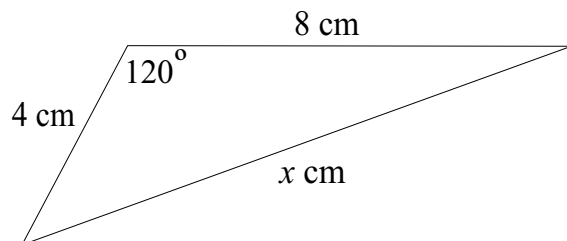
Start a new booklet for each question.

**QUESTION ELEVEN** (15 marks) Use a separate writing booklet.**Marks**

- (a) Write  $\frac{\sqrt{7}}{\sqrt{7}+2}$  with a rational denominator. **2**
- (b) Simplify:
- (i)  $\frac{\sin \theta}{\cos \theta}$  **1**
- (ii)  $1 - \sin^2 \theta$  **1**
- (c) Find the equation of the line with  $x$ -intercept 4 and  $y$ -intercept 2. **2**
- (d) Calculate the perpendicular distance from the point  $(-5, -7)$  to the line  $4x - 3y - 6 = 0$ . **2**
- (e) Differentiate:
- (i)  $e^{5+10x}$  **1**
- (ii)  $(3x - 2)^3$  **1**
- (iii)  $x \cos x$  **2**
- (f) Find:
- (i)  $\int \frac{1}{x^2} dx$  **1**
- (ii)  $\int \cos 7x dx$  **1**
- (iii)  $\int \frac{3}{3x+1} dx$  **1**

**QUESTION TWELVE** (15 marks) Use a separate writing booklet.**Marks**

(a)

**2**

Use the cosine rule in the diagram above to find  $x$  as a surd in simplest form.

(b) (i) Factorise  $2x^2 - 3x - 2$ .**1**(ii) Sketch the parabola  $y = 2x^2 - 3x - 2$  showing its  $x$  intercepts.**1**(iii) Solve  $2x^2 - 3x - 2 < 0$ .**1**

(c) Consider the finite sequence 24, 37, 50, ..., 7070.

(i) Find the 79th term.

**1**

(ii) Determine the number of terms in the sequence.

**2**(d) Find the gradient of the normal to the curve  $y = 4\sqrt{x}$  at the point (9, 12).**3**(e) The parabola  $\mathcal{P}$  has equation  $y^2 = 8(x + 2)$ .(i) Write down the coordinates of the vertex of  $\mathcal{P}$ .**1**(ii) Find the coordinates of the focus of  $\mathcal{P}$ .**1**(iii) Write down the equation of the directrix of  $\mathcal{P}$ .**1**(iv) Sketch the parabola  $\mathcal{P}$  showing the features found above, as well as any  $x$  or  $y$  intercepts.**1**

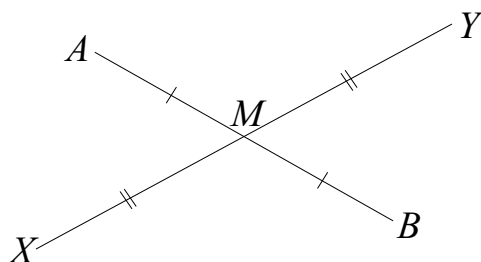
**QUESTION THIRTEEN** (15 marks) Use a separate writing booklet.**Marks**

- (a) A fuel tank ruptured and as a consequence the fuel began to leak out. The volume  $V$  litres of fuel remaining in the tank  $t$  minutes after the rupture was given by

$$V = 2000 - 100t - 10t^2.$$

- (i) How much fuel was in the tank when the rupture occurred? 1
- (ii) What percentage of the fuel remained in the tank after 4 minutes? 1
- (iii) At what rate was the fuel leaking out after 4 minutes? 1
- (iv) How long did it take for the tank to empty? 2

(b)



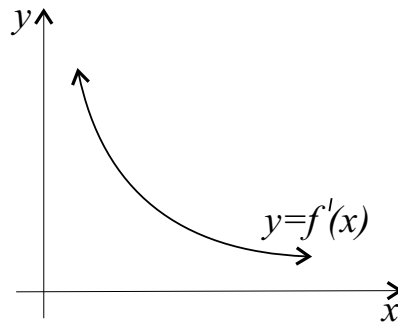
The diagram above shows two intervals  $AB$  and  $XY$  that bisect each other at  $M$ .

- (i) Prove that the triangles  $AMX$  and  $BM Y$  are congruent. 2
- (ii) Hence prove that  $AX$  and  $BY$  are parallel. 2
- (c) At the start of the year 2000 there were 40 foxes on McLean's cattle station. Since then the number of foxes  $F$  has increased exponentially according to the equation  $F = Ae^{kt}$ , where  $A$  and  $k$  are constants and  $t$  is the number of years since the start of 2000. The fox population had reached 120 at the start of 2008.
- (i) What is the value of  $A$ ? 1
- (ii) Find the exact value of  $k$ . 2
- (iii) What was the fox population at the start of 2015? 1
- (iv) During which year will the fox population reach 1000? (Assume that the population continues to grow exponentially according to the given equation.) 2

**QUESTION FOURTEEN** (15 marks) Use a separate writing booklet.

**Marks**

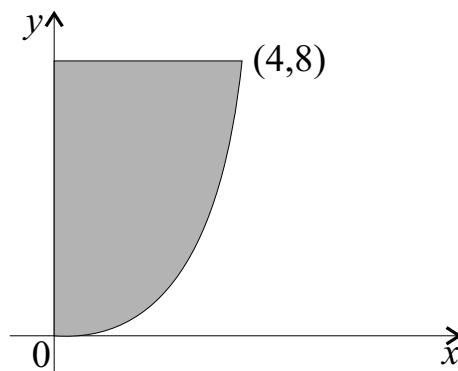
(a)



The diagram above shows the graph of the gradient function  $y = f'(x)$  of the curve  $y = f(x)$ .

- (i) State whether the curve  $y = f(x)$  is increasing or decreasing throughout its domain. 1
- (ii) State whether the curve  $y = f(x)$  is concave up or concave down throughout its domain. 1

(b)



The shaded region in the diagram above is bounded by the curve  $y = \frac{1}{8}x^3$ , the  $y$ -axis and the line  $y = 8$ .

- (i) Make  $x$  the subject of the equation  $y = \frac{1}{8}x^3$ . 1
  - (ii) Hence, or otherwise, find the area of the region. 2
  - (iii) Find the volume of the solid formed when the region is rotated about the  $y$ -axis. 2
- (c) A special commemorative tree in Canberra was 2 metres tall when it was planted many years ago. It grew 90 cm in the first year, and since then each year's growth has been 70% of the previous year's growth.
- (i) By how much, correct to the nearest centimetre, did the tree grow in its fifth year? 2
  - (ii) What is the limiting height of the tree? 2
- (d) The line  $y = mx - 2$  is a tangent to the circle  $x^2 + y^2 = 1$ .
- (i) By solving simultaneously, show that  $(1 + m^2)x^2 - 4mx + 3 = 0$ . 2
  - (ii) Use the discriminant of the quadratic equation in part (i) to find the two possible values of  $m$ . 2

**Exam continues overleaf ...**

**QUESTION FIFTEEN** (15 marks) Use a separate writing booklet.**Marks**

- (a) Consider the curve with equation
- $y = \frac{1}{3}x^3 - 3x^2 + 11x - 9$
- .

(i) Show that the curve has no stationary points.

**2**(ii) Show that there is a point of inflexion at  $(3, 6)$ .**1**(iii) How many  $x$  intercepts does the curve have? Justify your answer.**1**

- (b) A car is travelling along a straight horizontal road in outback Australia, with velocity
- $v$
- km/h given by

$$v = 100 + 20e^{-0.2t}.$$

Suppose that the car has displacement  $x = 0$  when  $t = 0$ .(i) Write down a formula for the acceleration of the car after  $t$  hours, and hence state whether the car is speeding up or slowing down.**2**(ii) Find a formula for the displacement of the car after  $t$  hours.**2**

(iii) How far, to the nearest kilometre, does the car travel in the third hour?

**1**

(iv) What is the limiting velocity of the car?

**1**

(v) Sketch the velocity-time graph.

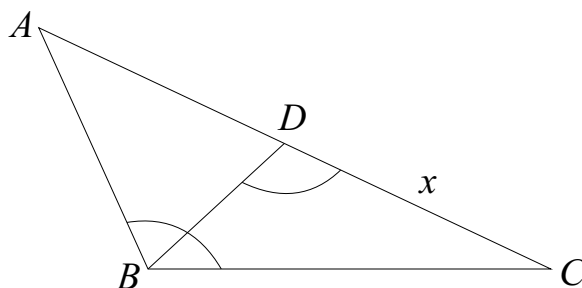
**1**

- (c) Michael is about to retire and has arranged for his superannuation lump sum of
- $\$P$
- to be deposited into an account which earns 3% per annum interest, with interest paid monthly on the balance. He intends to withdraw
- $\$6400$
- at the end of each month to live off.

Suppose that  $\$A_n$  is the balance immediately after the  $n$ th withdrawal.(i) Use the fact that  $A_1 = P(1.0025) - 6400$  to determine an expression for  $A_2$ .**1**(ii) Hence show that  $A_3 = P(1.0025)^3 - 6400(1.0025)^2 - 6400(1.0025) - 6400$ .**1**(iii) Find the value of  $P$  to the nearest integer if the balance will be zero after the 180th withdrawal.**2**

**QUESTION SIXTEEN** (15 marks) Use a separate writing booklet.**Marks**

(a)



The diagram above shows a triangle  $ABC$ . The point  $D$  is chosen on  $AC$  so that  $\angle BDC = \angle ABC$ .

(i) Prove that the triangles  $BDC$  and  $ABC$  are similar.

1

(ii) Given that  $DC = x$  and  $3DC = 2BC$ , find  $AD$  in terms of  $x$ .

3

(b) An open cylindrical water tank has base radius  $x$  metres and height  $h$  metres. Each square metre of the base costs  $a$  dollars to manufacture and each square metre of the curved surface costs  $b$  dollars, where  $a$  and  $b$  are constants. The combined cost of the base and curved surface is  $c$  dollars.

(i) Find  $c$  in terms of  $a$ ,  $b$ ,  $x$  and  $h$ . (Note that the curved surface has area  $2\pi xh$ .)

1

(ii) Show that the volume  $V$  of the tank in cubic metres is given by

2

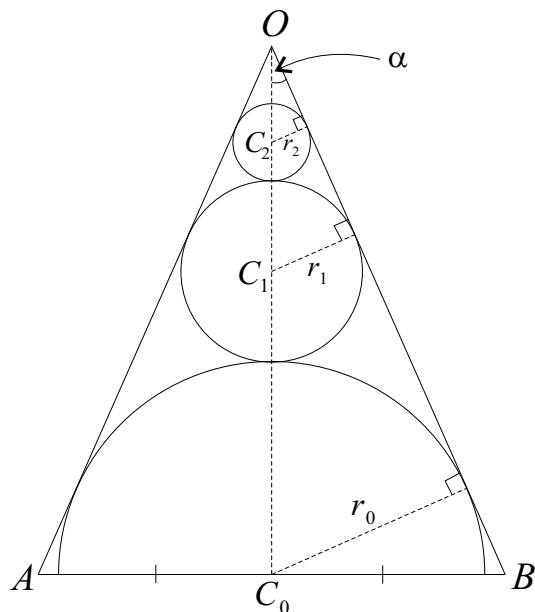
$$V = \frac{x}{2b} (c - \pi ax^2).$$

(iii) If  $x$  can vary, prove that  $V$  is maximised when the cost of the base is  $\frac{c}{3}$  dollars.

4

**QUESTION SIXTEEN** (Continued)

(c)



In the diagram above  $\triangle OAB$  is isosceles, with  $OA = OB$  and  $\angle AOB = 2\alpha$ . The semicircle of radius  $r_0$  has centre  $C_0$ , the midpoint of  $AB$ , and the sides  $OA$  and  $OB$  are tangents. Infinitely many circles are stacked on top of the semicircle so that each circle is tangent to the circles immediately above and below it, and also tangent to  $OA$  and  $OB$ . The circles have radii  $r_1, r_2, r_3, \dots$  and centres  $C_1, C_2, C_3, \dots$ . The semicircle and the first two circles are shown in the diagram.

- (i) Write down an infinite series for  $OC_0$  in terms of  $r_0, r_1, r_2, \dots$  and write down a similar series for  $OC_1$ . 1
- (ii) Prove that  $\frac{r_1}{r_0} = \frac{1 - \sin \alpha}{1 + \sin \alpha}$ . 3

\_\_\_\_\_ End of Section II \_\_\_\_\_

**END OF EXAMINATION**

(1)  $\frac{r}{p}$

(B)

(2)  $y \neq -4$

(A)

(3)  $(n+1)x^n$

(C)

(4)  $f(a) > 0$  and  $f'(a) < 0$

(B)

(5)  $\frac{1}{2}$  and  $\frac{1}{4}$

(C)

(6)  $\log 2$

(A)

(7)  $\frac{11}{3}$

(D)

(8)

(B)

(9)  $\int_{-a}^a f(x) dx \neq 2 \int_0^a f(x) dx$

(C)

(10)  $\sqrt{x^2} \neq x$

(D)

ONE MARK EACH

(11)(a)  $\frac{\sqrt{7}}{\sqrt{7}+2} \times \frac{\sqrt{7}-2}{\sqrt{7}-2}$

$$= \frac{7-2\sqrt{7}}{3}$$

(b)(i)  $\frac{\sin \theta}{\cos \theta} = \tan \theta$

(ii)  $1 - \sin^2 \theta = \cos^2 \theta$

(c) Line through (4,0) and (0,2).

$$m = \frac{2-0}{0-4}$$

$$= -\frac{1}{2}$$

The line has equation

$$y = -\frac{1}{2}x + 2$$

(or  $x + 2y - 4 = 0$ )

(d)  $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$$= \frac{|-20 + 21 - 6|}{5}$$

$$= 1 \text{ unit}$$

(e)(i)  $\frac{d}{dx}(e^{5+10x}) = 10e^{5+10x}$

(ii)  $\frac{d}{dx}(3x-2)^3 = 9(3x-2)^2$

(iii) By the product rule

$$\frac{d}{dx}(x \cos x) = \cos x - x \sin x$$

(f)(i)  $\int x^{-2} dx = \frac{x^{-1}}{-1} + c$   
$$= -\frac{1}{x} + c$$

(ii)  $\int \cos 7x dx = \frac{1}{7} \sin 7x + c$

(iii)  $\int \frac{3}{3x+1} dx = \log_e(3x+1) + c$

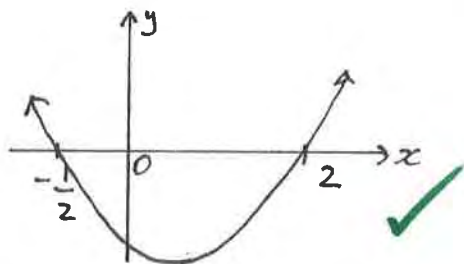
(No penalty for omission of 'c'.)

$$\begin{aligned}
 (12)(a) \quad x^2 &= 4^2 + 8^2 - 2 \times 4 \times 8 \times \cos 120^\circ \\
 &= 64 + 16 - 64 \times -\frac{1}{2} \\
 &= 80 + 32 \\
 &= 112
 \end{aligned}$$

$$\begin{aligned}
 \text{so } x &= \sqrt{112} \\
 &= 4\sqrt{7}
 \end{aligned}$$

$$(b)(i) \quad 2x^2 - 3x - 2 = (2x+1)(x-2)$$

(ii)



$$(iii) \quad -\frac{1}{2} < x < 2$$

(c) The sequence is arithmetic with  $a = 24$  and  $d = 13$ .

$$\begin{aligned}
 (i) \quad T_{79} &= a + 78d \\
 &= 24 + 78(13) \\
 &= 1038
 \end{aligned}$$

(ii) Let  $T_n = 7070$  and solve for  $n$ .

$$24 + (n-1)(13) = 7070$$

$$13n + 11 = 7070$$

$$n = 543$$

So there are 543 terms.

$$\begin{aligned}
 (d) \quad y &= 4x^{\frac{1}{2}} \\
 \text{so } y' &= 2x^{-\frac{1}{2}} \\
 &= \frac{2}{\sqrt{x}}
 \end{aligned}$$

$$\text{When } x = 9, \quad y' = \frac{2}{3}$$

so the normal has gradient  $-\frac{3}{2}$ .

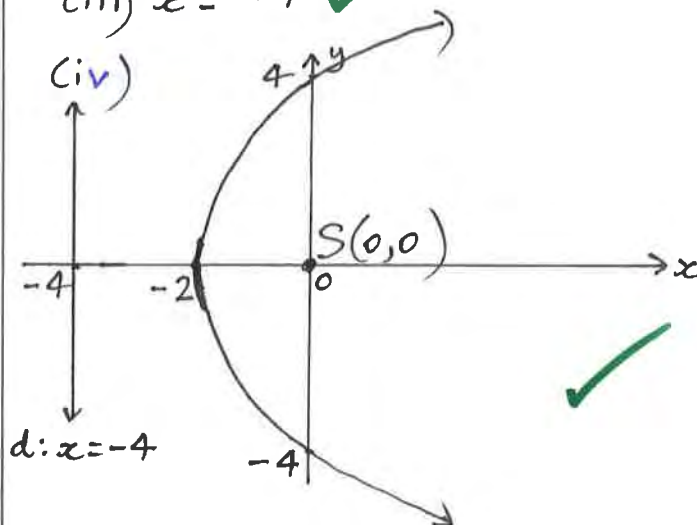
(e)(i) The parabola is concave right with vertex  $(-2, 0)$ .

(ii) Also  $4a = 8$  so  $a = 2$ .

So the focus is  $(0, 0)$ .

$$(iii) \quad x = -4$$

(iv)



When  $x = 0$ ,

$$y^2 = 16$$

$y = \pm 4$  (y-intercepts)

(13)(a)(i) When  $t=0$ ,  
 $V = 2000 \text{ L}$  ✓

(ii) When  $t=4$ ,  
 $V = 2000 - 400 - 160$   
 $= 1440$

so percentage remaining is

$$\frac{1440}{2000} \times \frac{100}{1} \% = 72\% \text{ ✓}$$

(iii)  $\frac{dV}{dt} = -100 - 20t$

When  $t=4$ ,

$$\frac{dV}{dt} = -100 - 80$$

$$= -180$$

So the fuel was leaking  
out at  $180 \text{ L/min}$ . ✓

(iv) When  $V=0$ ,

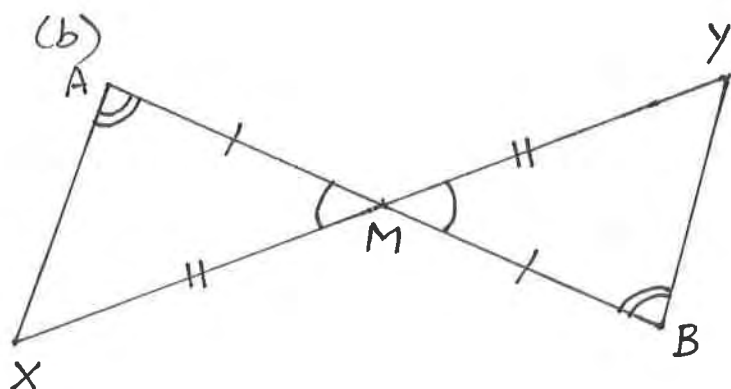
$$0 = 2000 - 100t - 10t^2$$

$$t^2 + 10t - 200 = 0$$

$$(t+20)(t-10) = 0 \text{ ✓}$$

$$t > 0 \text{ so } t=10$$

so the tank took 10 min to empty. ✓



(i) In  $\Delta s$  AMX and BMY:

$$AM = BM \text{ (given)}$$

$$\angle AMX = \angle BMY \text{ (vertically opposite)}$$

$$XM = YM \text{ (given)}$$

$$\text{so } \Delta AMX \equiv \Delta BMY \text{ (S.A.S.)} \text{ ✓}$$

(ii)

$$\angle XAM = \angle YBM \text{ (matching angles of congruent triangles)} \text{ ✓}$$

$$\text{so } AX \parallel YB \text{ (alternate angles are equal)} \text{ ✓}$$

(c)(i) When  $t=0$ ,  $F=40$

$$\text{so } 40 = Ae^0$$

$$40 = A$$

$$A = 40 \text{ ✓}$$

(ii) When  $t=8$ ,  $F=120$

$$\text{so } 120 = 40e^{8k}$$

$$e^{8k} = 3 \text{ ✓}$$

$$8k = \log_e 3$$

$$k = \frac{1}{8} \ln 3 \text{ ✓}$$

(iii) When  $t=15$ ,

$$F = 40e^{15k}$$

$$= 313.806 \dots$$

$$\div 314 \text{ ✓}$$

(iv) When  $F=1000$ ,

$$1000 = 40e^{kt}$$

$$e^{kt} = 25 \text{ ✓}$$

$$kt = \log_e 25$$

$$t = \frac{\ln 25}{k}$$

$$= 23.439 \dots$$

so the fox population will  
reach 1000 during 2023. ✓

(14)(a) (i)  $f'(x) > 0$   
so  $f(x)$  is increasing.

(ii)  $f''(x) < 0$  so the curve is concave down.

(b)(i)  $y = \frac{1}{8}x^3$   
 $x^3 = 8y$   
 $x = 2y^{\frac{1}{3}}$

(ii) Area =  $\int_0^8 2y^{\frac{1}{3}} dy$   
 $= 2 \left[ \frac{3}{4} y^{\frac{4}{3}} \right]_0^8$   
 $= \frac{3}{2} (16 - 0)$   
 $= 24 \text{ units}^2$

OR Area =  $4 \times 8 - \int_0^4 \frac{1}{8} x^3 dx$   
 $= 32 - \frac{1}{8} \left[ \frac{x^4}{4} \right]_0^4$   
 $= 32 - \frac{1}{8} (64)$   
 $= 32 - 8$   
 $= 24 \text{ units}^2$

(iii) Volume =  $\pi \int_0^8 x^2 dy$   
 $= \pi \int_0^8 4y^{\frac{2}{3}} dy$   
 $= 4\pi \left[ \frac{3}{5} y^{\frac{5}{3}} \right]_0^8$   
 $= \frac{12\pi}{5} (32 - 0)$   
 $= \frac{384\pi}{5} \text{ units}^3$

(4)

(c)(i) The annual growths form a GP with  $a = 90$  and  $r = 0.7$ .

$T_5 = ar^4$   
 $= 90(0.7)^4$   
 $= 21.609$   
 $\div 22 \text{ cm}$

(ii)  $S_{\infty} = \frac{a}{1-r}$   
 $= \frac{90}{0.3}$   
 $= 300 \text{ cm}$   
 $= 3 \text{ m}$

So the limiting height is 5 m.

(d)(i)  $y = mx - 2$  (1)  
 $x^2 + y^2 = 1$  (2)

Substitute (1) into (2):

$x^2 + (mx - 2)^2 = 1$   
 $x^2 + m^2x^2 - 4mx + 4 = 1$   
 $(1+m^2)x^2 - 4mx + 3 = 0$

(ii)  $\Delta = b^2 - 4ac$   
 $= (-4m)^2 - 4(3)(1+m^2)$   
 $= 16m^2 - 12 - 12m^2$   
 $= 4m^2 - 12$   
 $= 4(m^2 - 3)$

Since the line is a tangent there is only one point of intersection.

So  $\Delta = 0$ .

So  $m^2 - 3 = 0$

$m^2 = 3$

$m = \pm\sqrt{3}$

$$(15)(a)(i) y = \frac{1}{3}x^3 - 3x^2 + 11x - 9$$

$$\begin{aligned} \text{so } y' &= x^2 - 6x + 11 \\ &= (x^2 - 6x + 9) + 2 \\ &= (x-3)^2 + 2 \\ &> 2 \text{ for all } x \end{aligned}$$

so  $y'$  is always positive,  
so  $y'$  is never zero,  
so there are no stationary points.

$$(ii) y'' = 2x - 6$$

$$\text{So } y'' = 0 \text{ at } x = 3.$$

We must show that the concavity changes at  $x = 3$ .

| $x$       | 2    | 3 | 4  |
|-----------|------|---|----|
| $y''$     | -2   | 0 | +2 |
| CONCAVITY | down |   | up |

When  $x = 3$ ,

$$y = 9 - 27 + 33 - 9 = 6.$$

So  $(3, 6)$  is a point of inflexion.

(iii) The curve has ONE  $x$ -intercept because it is strictly increasing for all values of  $x$ .

$$(b)(i) v = \dot{x} = 100 + 20e^{-0.2t}$$

$$\text{so } \ddot{x} = -4e^{-0.2t}$$

which is negative for all  $t > 0$ .

But  $v$  is positive for all  $t > 0$

so the car is slowing down.

$$(ii) x = \int (100 + 20e^{-0.2t}) dt$$

$$= 100t - 100e^{-0.2t} + c$$

$$\text{When } t = 0, x = 0$$

$$\text{so } c = 100$$

$$\begin{aligned} \text{so } x &= 100t - 100e^{-0.2t} + 100 \\ &= 100(t - e^{-0.2t} + 1) \end{aligned}$$

$$(iii) \text{When } t = 2, x = 200 - 100e^{-0.4} + 100 = 232.967...$$

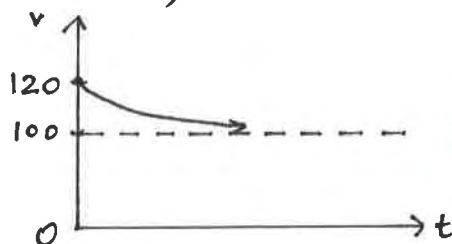
$$\text{When } t = 3, x = 300 - 100e^{-0.6} + 100 = 345.118...$$

So the car has travelled 112 km (to the nearest km).

$$(iv) \text{As } t \rightarrow \infty, e^{-0.2t} \rightarrow 0,$$

$$\text{so } v \rightarrow 100 \text{ km/h.}$$

$$(v) \text{When } t = 0, v = 120.$$



$$(c) 3\% \text{ p.a.} = 0.25\% \text{ per month}$$

$$\text{so } r = 0.0025$$

$$(i) A_2 = A_1(1.0025) - 6400 = P(1.0025)^2 - 6400(1.0025) - 6400$$

$$(ii) A_3 = A_2(1.0025) - 6400 = P(1.0025)^3 - 6400(1.0025)^2 - 6400(1.0025) - 6400$$

$$(iii)$$

$$A_n = P(1.0025)^n - 6400(1 + 1.0025 + \dots + 1.0025^{n-1})$$

$$A_{180} = 0$$

$$\text{so } P(1.0025)^{180} = \frac{6400(1.0025^{180} - 1)}{0.0025}$$

$$P = \frac{6400(1.0025^{180} - 1)}{(1.0025)^{180}(0.0025)}$$

$$= \$926\,755 \text{ (to the nearest dollar)}$$

(16)(a)(i) In  $\Delta BDC$  and  $ABC$ :

$\angle BDC = \angle ABC$  (given)  
 $\angle BCD = \angle ACB$  (common)

so  $\Delta BDC \parallel \Delta ABC$  (A.A.)

(ii)  $\frac{BC}{AC} = \frac{DC}{BC}$  (matching sides in the same ratio)

so  $BC^2 = AC \times DC$

so  $\left(\frac{3x}{2}\right)^2 = AC \times x$

so  $AC = \frac{9x}{4}$

so  $AD = \frac{9x}{4} - x$   
 $= \frac{5x}{4}$

(b)(i)  $c = \pi x^2 a + 2\pi x h b$

(ii)  $V = \pi x^2 h$   
 From (i)  $h = \frac{c - \pi x^2 a}{2\pi x b}$

so  $V = \frac{\pi x^2}{2\pi x b} (c - \pi x^2 a)$   
 $= \frac{x}{2b} (c - \pi x^2 a)$

(iii)  $V = \frac{cx}{2b} - \frac{\pi ax^3}{2b}$

so  $V' = \frac{c}{2b} - \frac{3\pi ax^2}{2b}$

Let  $V' = 0$ .

Then  $x^2 = \frac{c}{2b} \cdot \frac{2b}{3\pi a}$   
 $= \frac{c}{3\pi a}$

Since  $x > 0$ , the only stationary point under consideration is at

$x = \left(\frac{c}{3\pi a}\right)^{\frac{1}{2}}$

$V'' = -\frac{3\pi ax}{b} < 0$  when  $x > 0$ .

So  $V$  is maximised when  $x = \left(\frac{c}{3\pi a}\right)^{\frac{1}{2}}$ .

The cost of the base is  $\pi ax^2$ .

So when  $x^2 = \frac{c}{3\pi a}$ , the cost is

$\pi a \cdot \frac{c}{3\pi a} = \frac{c}{3}$  dollars.

(c)(i)  $OC_0 = r_0 + 2r_1 + 2r_2 + \dots$   
 and  $OC_1 = r_1 + 2r_2 + 2r_3 + \dots$

(ii) It follows from (i) that

$OC_0 - OC_1 = (r_0 + 2r_1) - r_1$   
 $= r_0 + r_1$  (A)

From the diagram,

$\sin \alpha = \frac{r_0}{OC_0} = \frac{r_1}{OC_1}$

so  $OC_0 = \frac{r_0}{\sin \alpha}$

and  $OC_1 = \frac{r_1}{\sin \alpha}$  (B)

Substituting from (B) into (A):

$\frac{r_0}{\sin \alpha} - \frac{r_1}{\sin \alpha} = r_0 + r_1$

$r_0 - r_1 = r_0 \sin \alpha + r_1 \sin \alpha$

$r_0 - r_0 \sin \alpha = r_1 + r_1 \sin \alpha$

$r_0(1 - \sin \alpha) = r_1(1 + \sin \alpha)$

so  $\frac{r_1}{r_0} = \frac{1 - \sin \alpha}{1 + \sin \alpha}$